

GLACIS FIELD GUIDE

The offline penetration-testing worksheet & practical-exam companion

Version 0.2.0 “Pulse” · eJPTv2 / eCPPT friendly · works the same in labs, CTFs and client work

GLACIS is a keyboard-driven terminal worksheet that sits next to your tools and remembers everything for you: targets, ports, services, credentials, flags, proofs, dead ends and the commands you ran. It **never attacks anything itself** — it shows you the right command, copies it to your clipboard, and you run it in your own terminal. Everything lives in a local SQLite file. No network, no cloud, no AI in the loop.

The whole method in one line

SCAN	›	RECORD	›	COPY	›	RUN	›	MARK	›	EXPORT
Why it wins in practical exams • One screen answers: which host, what's open, what's next, what did I prove? • Ready-made commands per service — press Enter to copy, zero typos. • Credential matrix stops password-reuse amnesia across machines. • Pulse dashboard shows exactly which machine is being neglected.					Exam-integrity compliance • 100% offline — local SQLite, zero network calls, zero telemetry. • Zero autonomous action — GLACIS copies commands; you execute them. • No cloud assistance — nothing that violates exam rules. • Equivalent to permitted personal notes (Obsidian / CherryTree), just faster.					

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1 Start in 10 minutes

Do this once per lab. The first launch opens a three-step quick-start card — press any key to dive in, and `:welcome` brings it back anytime. Empty panels always tell you which key fills them, so you can explore without this guide in hand.

#	Step	Do exactly this
1	Install & launch	<code>pip install -e . # inside the repo</code> <code>glacis # opens the TUI</code>
2	Make it yours	<code>T # theme picker, d = save default</code> <code>:theme midnight # or direct switch</code>
3	Create the lab workspace	<code>W # workspace manager</code> <code>:ws init lab01 # scaffolds scans/ enum/</code> <code># screenshots/ notes/ loot/</code>
4	Get targets in fast	<code>I # import nmap XML/txt/gnmap</code> <code>:t 10.10.10.20 # or add one by hand</code> <code>:subnet 10.10.10.20 10.10.10.0/24</code>
5	Load a methodology	<code>m # template picker</code> <code>:m ejpt # master exam checklist</code>
6	Work the loop (§3)	<code>j/k, Enter, Space # copy cmd, run it,</code> <code># mark result, repeat</code>
7	Export & protect	<code>:export exam # markdown dossier</code> <code>glacis export --format html -o report.html</code> <code>glacis backup --label checkpoint</code>

Prefer plain shell? Every TUI action has a CLI twin — see the pocket reference in §10.

■ Mind-set

GLACIS is the co-pilot, not the pilot. It remembers, suggests syntax and tracks progress; **you** scan, exploit and decide. Write down everything the moment you learn it — future-you at hour six is a stranger.

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The five stations

GLACIS splits the work across five screens. Switch with the number keys — anything you record is shared by all of them.

Key	Station	What it gives you	Go here when...
0	Pulse — engagement dashboard	Stat cards, 14-day momentum sparkline, per-target scorecards (A–F), next-action queue, live timeline of everything recorded.	You finish a box, feel lost, or wonder what to touch next.
1	Cockpit — the workbench	Target tree, services & ports with triage states, methodology checklist, notes and the command console.	Actually working a machine — this is home.
2	Playbooks	Offline command encyclopedia: per-service attack recipes you can copy with Enter.	You know the service, not the syntax.
3	Credentials	2D matrix: accounts × services. Compiles spray commands; tracks valid / pwned / invalid.	You hold creds and need to know where they work.
4	Loot & Flags	User/root flags, foothold & privesc proofs, question evidence, disk-loot browser, rabbit-hole log.	You captured something — or you're stuck.

■ New in v0.2.0 — Pulse, HTML reports and snapshots

Press **0** anytime for the new **Pulse** dashboard: momentum sparkline, per-target scorecards and a next-action queue built from your own open items. Export a self-contained **HTML report** with `glacis export --format html`, and protect long sessions with `glacis backup`. All still 100% offline — see §9.

3 The core loop — six beats, every machine

Everything in GLACIS serves this loop. It takes seconds per beat and leaves a perfect trail: **what you tried, what worked, what to prove.**

Beat	Key / command	What happens
1 • RECORD	:t 10.10.10.20 · :s 445/tcp smb	Target and services land in the cockpit; checklists and playbooks key off them.
2 • COPY	j / k Enter	Highlight a service — the console shows a ready command with the IP substituted. Enter copies it.
3 • RUN	(your terminal / tmux pane)	Paste and execute. GLACIS never runs anything.
4 • MARK	Space	Cycle the service status: UNTESTED → CHECKED → DEAD-END → DEFERRED. The checklist % and Pulse coverage update.
5 • CAPTURE	:c user:pass · :uflag ... · :q 7 ...	Creds, flags and question proofs go straight into the ledger while you have them.
6 • PIVOT	]	Next target. Check station 0 if you're unsure what "next" is.



Station 1 Cockpit — attack-surface tree (left), services with triage states (top right), methodology & notes (bottom), command console underneath.

■ PRO-TIP — the zero-mouse flow

Highlight a port with **j / k** → read the console → press **.** to swap tool recipes (whatweb → feroxbuster → gobuster → nikto → curl) → **Enter** copies → run it in tmux → **Space** to mark the result. Your hands never leave the keyboard.

4 Keyboard map — grouped by job

Single letters fire instantly unless a modal or the console input is focused. On day one you only need the eight keys in the blue box — the rest come naturally.

A · Survival — learn these first

Key	Does
0 1 2 3 4	Jump to Pulse / Cockpit / Playbooks / Credentials / Loot & Flags.
j / k	Move down / up in the focused list or tree (arrows work too).
Enter	Copy the highlighted command / value (matrix: compiled spray command).
Space	Cycle triage status; in station 3 cycle UNTESTED → VALID → PWN3D → INVALID; in station 4 preview loot.
y	Quick-copy the row's core value (IP, IP:port, secret, note text).
E	Exam Mode — persistent EXAM MODE · OFFLINE NOTES badge in the header (proctored exams).
/ or Ctrl+F	Global search across everything; Enter copies the top hit.
?	Full help & shortcut reference.
q	Quit (your data is already saved).

B · Capture — get findings in fast

Key	Does
t / s / c / n	Add target / service / credential / note via modal (or just type :t, :s, :c, :n in the console).
f	Record a finding with optional severity.
K	Add a custom checklist item (capital K — lowercase k moves).
m	Methodology template picker.
I	Import a scan: Nmap XML/text/gnmap, FFUF, feroxbuster, gobuster — staged for review before anything is committed.
g	Record captured user/root flags.
a	Link a question number to proof evidence (station 4).
v	Paste clipboard screenshot into screenshots/ and attach as evidence.

C · Move fast

Key	Does
[/]	Previous / next target — rotate machines in one keystroke.
, / .	Cycle the tool recipe for the highlighted service in the console.
w · h / l	Cycle panel focus; jump to left / right column.
b · z	Collapse sidebar · zoom focused panel full-screen.
o · r · T	Toggle scope · offline reference modal · theme picker (d saves; G toggles derived guidance, off by default).

■ Day-one eight

0-4 stations · **j / k** move · **Enter** copy · **Space** mark · **:** console · **y** copy value · **/** search · **?** help. Everything else can wait until you feel the rhythm — and **r** opens the offline syntax cheatsheet.

5 Console command reference

Press **:** and type. Tab completes; Esc returns to the workbench.

Capture

Command	Effect
<code>:t 10.10.10.20 [hostname] [os] [subnet] [pivot]</code>	Add target (also: <code>add target ...</code> in plain words).
<code>:s 445/tcp smb [-- notes]</code>	Record an open service on the active target.
<code>:c admin:Secret123! [scope]</code>	Record a credential; scope tags the service.
<code>:c crack 3 Password123!</code>	Upgrade a captured hash (row 3) to plaintext in place.
<code>:n any free text</code>	Timestamped note attached to the active target.
<code>:f Suspicious ACL on /admin</code>	Finding with optional severity prompt.

Knowledge & navigation

Command	Effect
<code>:ref winrm · :ref smb</code>	Offline command reference for a service/tool.
<code>:m ejpt · :m web append</code>	Load methodology template (replace or append).
<code>:theme midnight</code>	Live theme switch (persist with T → d).

Host state, proofs & evidence

Command	Effect
<code>:uflag 7a9e... · :rflag f04b...</code>	Record user / root flag for the active target.
<code>:foothold smb null session · :privesc SUID ...</code>	Record how you got in and how you went up.
<code>:q 7 /etc/shadow root hash</code>	Pin exam-question 7 to its proof string.
<code>:stuck wp-login brute = rabbit hole</code>	Log a rabbit hole (also <code>:dead</code>).
<code>:clue backup.zip in downloads share</code>	Log the breakthrough clue.
<code>:ev latest [desc] · :paste-ev [desc]</code>	Attach newest OS screenshot / paste clipboard image as evidence.
<code>:subnet 10.10.10.20 10.10.10.0/24</code>	Tag which subnet a target lives in.
<code>:pivot 10.10.10.20 172.16.1.0/24</code>	Mark a dual-homed pivot host + its hidden subnet.

Workspace & output

Command	Effect
<code>:ws lab02 · :ws init lab02</code>	Switch workspace / scaffold a new lab folder tree.
<code>:lhost auto · :lport 4444</code>	Attacker VPN IP (auto-detects tun0/wg0) & listener port — substituted into playbooks.
<code>:w rockyou · :w common · :w medium</code>	Copy standard wordlist paths to clipboard (see aliases below).
<code>:export wordlists</code>	Write deduplicated loot/users.txt + loot/passwords.txt.
<code>:export exam</code>	Markdown dossier: targets, question proofs, flags, creds, audit trail.
<code>:w <alias></code>	Copy a wordlist path: <code>rockyou · common · medium · big / small · raft-d / raft-f · users / passwords · fasttrack</code> .

6

Workflow I — first contact with a machine

The repeatable 10-minute pattern for any new box. Left: what you run in your terminal. Right: what you tell GLACIS.

#	In your terminal	In GLACIS
1	<code>nmap -sn 10.10.10.0/24</code>	Spot the alive hosts.
2	—	<code>:t 10.10.10.20 hostname os · :subnet ...</code>
3	<code>nmap -sS -p- --min-rate 1000 10.10.10.20</code>	— (or import whole XML next step)
4	<code>nmap -sV -sC -p 22,80,445 10.10.10.20 -oX out.xml</code>	I → review staged hosts/services → commit
5	—	<code>:m ejpt # methodology on this target</code>
6	run the copied recipe	j/k highlight service, · pick tool, Enter copy
7	(paste & execute)	Space
8	find creds / flag / foothold	<code>:c ... · :uflag ... · :foothold ... · :q ...</code>
9	stuck? move on	<code>:stuck ... then] # rotate to next target</code>

II

Workflow II — the exam battle plan (5 phases)

Phase	Goal	GLACIS verbs
PHASE 1 · Scope & discovery	Find every machine and put it on the record.	<code>nmap -sn / arp-scan → :t + :subnet per host · :pivot for dual-homed boxes</code>
PHASE 2 · Enumerate everything	Full TCP sweep, then versions; import instead of typing.	<code>nmap -p- → -sV -sC -oX → I import · Space-mark each service as you clear it</code>
PHASE 3 · Low-hanging fruit	Anonymous & default access first, then web fuzzing.	<code>ftp/smb/snmp quick wins → :w common + feroxbuster → :f findings with severity</code>
PHASE 4 · Sprawl with creds	Every cred into the matrix; spray SMB/SSH/WinRM everywhere (\$7).	<code>:c harvest → station 3 → Enter = spray cmd → Space = VALID/PWN3D</code>
PHASE 5 · Prove & submit	Flags, question proofs, clean dossier — before you forget.	<code>:uflag/:rflag · :q n proof · :export exam · glacis backup --label submitted</code>

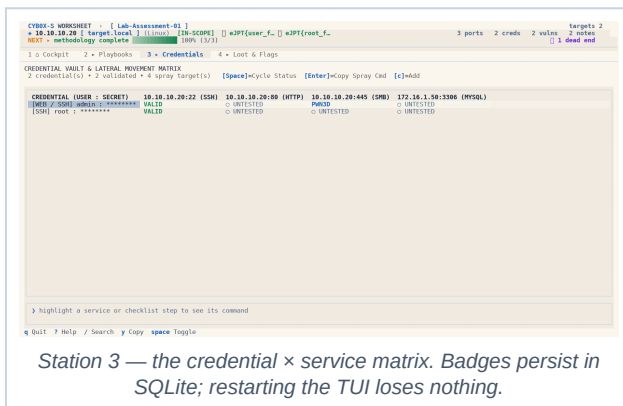
■ THE 20-MINUTE RULE — the single highest-value habit

Stuck on one service for 20 minutes with no new lead? **Stop.** **Space** it to DEFERRED, type **:stuck what-you-tried**, press **]** and rotate. The Pulse queue and your DEAD-END trail will bring you back with fresh eyes. Most failed practicals are one 3-hour rabbit hole deep.

III Workflow III — the credential spray loop

Multi-machine exams are won on password reuse. Station 3 lays every credential against every service, so “did I try this password on this box?” always has a visible answer.

Step	Action
1 · HARVEST	Pull creds from web configs, DB dumps, SAM hive, shell history.
2 · RECORD	c :c dbuser:S3cret! mysql
3 · SPRAY	Station 3 → highlight a cred × service cell → Enter copies the compiled spray command.
4 · RUN & MARK	Run it; Space the cell: UNTESTED → VALID → PWN3D → INVALID (persisted).
5 · EXPLOIT THE WIN	PWN3D cell = your foothold on that box: back to station 1, capture flags, repeat.



■ Cracked a hash mid-session?

Don't duplicate the row — upgrade it: **:c crack <row> <plaintext>**. Then **:export wordlists** writes deduplicated loot/users.txt + loot/passwords.txt for hydra / netexec runs.

IV Workflow IV — pivot into the internal network

Step	Action
1 · SPOT & ROUTE	ip a on a pwned box → second NIC = hidden subnet. Record: :pivot 10.10.10.20 172.16.1.0/24 — the console switches to tunnel guidance.
2 · TUNNEL	chisel server on pwned box → client on Kali → proxychains nmap -sT 172.16.1.0/24
3 · RECORD THE REST	:t 172.16.1.50 --notes via-pivot — same core loop; the target tree shows both segments.
WHY IT PAYS	Exam questions ask about hidden subnets and routes , not just flags — and the dossier spells your routes out.

V

Workflow V — prove it and submit

Evidence you didn't write down is evidence you don't have. Station 4 is where proofs live.

You captured...	Do this
A user/root flag	:uflag <hash> · :rflag <hash>
An exam answer	:q <question#> <exact proof string> # or key a in station 4
A screenshot	V pastes the clipboard image into screenshots/ and links it as evidence · :ev latest desc
Loot on disk	The loot browser indexes scans/, enum/, loot/, screenshots/ — Space previews, Enter copies the path.
A dead end	:stuck ... / :clue ... # future-you says thanks



✓ The submission moment

Before touching the exam portal: **:export exam** → a self-contained Markdown dossier (target inventory, question proofs, flags, credential states, audit trail). Keep it open beside the portal and answer from it. Want something prettier to attach to a client report? **glacis export --format html -o report.html** renders a styled, self-contained HTML report (masked credentials, printable).

9

Pulse — see everything, choose smarter

Press **0**. Pulse is a read-only dashboard computed from what you already recorded: coverage (% services tested), methodology %, 14-day momentum sparkline, per-target scorecards with a transparent A–F grade, a **next-action queue** (your own untested services, missing proofs, open leads — ranked NOW / NEXT / LATER) and the newest timeline events.

Moment	What to look at
Just finished a box	Scorecards → the neglected machine (lowest coverage, no flags) is usually your next target.
Don't know what's next	The NOW/NEXT queue — it's your own TODOs, ordered. Work top down.
Feels like slow progress	Momentum sparkline — flat two days = rotate sooner, log rabbit holes.
CLI equivalent	<code>glacis stats · glacis timeline -n 40</code>

GLACIS WORKSHEET · [Lab-Assessment-01] · [Pulse]

• 10.10.10.20 [target.local] (Linux) [IN-SCOPE] T+0:00:00 [LHOST: 169.254.0.21:4444] [USER: √] [ROOT: √]

CHECKLIST · methodology complete

0 • Pulse 1 Cockpit 2 Playbooks 3 Credentials 4 Loot & Flags

• PULSE · Lab-Assessment-01 offline engagement intelligence

TARGETS SERVICES FINDINGS CREDs METHODOLOGY MOMENTUM 7D

2 4 2 2 100% 15

2 in scope 75% tested 1 high+ 1 evidence 3/3 steps 16 today

momentum - 14 days

TARGET SCORECARDS

target	os	grade	coverage	method	svc	find	flags
10.10.10.20 (target.local)	Linux	A	100%	100%	3	2	2
172.16.1.50 (db01.corp.internal)	Linux	F	0%	0%	1	0	0

NEXT ACTIONS — from your own open items

NEXT Test 3306/tcp MySQL · 172.16.1.50 never tested

TIMELINE — newest first (press 1-4 to work a station)

00-09 22:00 x Failure log · 172.16.1.50

00-09 22:00 x Failure log · 10.10.10.20

00-09 22:00 Evidence [screenshot] evidence/proof_screenshot_01.png · 10.10.10.20

00-09 22:00 Note · 10.10.10.20

00-09 22:00 Note · 10.10.10.20

00-09 22:00 Checklist [CHECKED] Pivoting & Route Discovery · 10.10.10.20

00-09 22:00 Checklist [CHECKED] SMB enumeration · 10.10.10.20

00-09 22:00 Checklist [CHECKED] TCP enumeration · 10.10.10.20

00-09 22:00 Credential root · 10.10.10.20

00-09 22:00 Credential admin · 10.10.10.20

STATION 0 · PULSE

PULSE · Live engagement dashboard · momentum · scorecards · next actions

STATUS · Ready for command execution or target exploration

[:] > Type command (it, is, ic, im, iw) or note... (i for menu, Tab to complete)

[w] panels [1-4] stations [?] help [q] quit

[0-4: Stations]

Station 0 Pulse — stat cards, momentum, scorecards, next actions and timeline. Zero inference: it only counts what you recorded.

10 Templates, CLI pocket reference & exam day

Methodology templates (:m <name>)

Template	Focus	Covers
:m ejpt	Master flow	Scope → discovery → enumeration → foothold → pivoting → privesc — the exam spine.
:m discovery	Network	Local subnets, ARP/ICMP sweeps, TTL guesses, dual-homed discovery.
:m web	Web app	Headers, robots, dir fuzzing, SQLi, LFI, XSS, command injection, uploads.
:m smb	SMB	Null sessions, share perms, backups in shares, enum4linux, RID cycling.
:m pivoting	Routing	Dual-homed detection, autoroute, SOCKS5, SSH tunnels, chisel, proxychains.
:m linux · :m windows	PrivEsc	SUID/SGID, sudo -l, cron, capabilities / Selmpersonate, unquoted paths, AIE.
:m ftp · :m ssh · :m snmp	Services	Anonymous login, banner CVEs, key perms / communities, MIB walk.
:m databases · :m cracking	Data & hashes	Blank root logins, LOAD_FILE, xp_cmdshell / hash ID, John, Hashcat modes, hydra.

CLI pocket reference — when you'd rather stay in the shell

Command	Records / does
glacis t 10.10.10.20 -h dc01 --os Linux	Add a target.
glacis s 10.10.10.20 445/tcp SMB -v 'Samba 4.3'	Add a service.
glacis c admin:secret --source backup.zip	Add a credential.
glacis n 'checked share, nothing' · glacis f 'anon SMB' -s HIGH	Note / finding.
glacis import scan.xml	Staged scan import (same review as key I).
glacis search 'backup'	Search everything.
glacis stats · glacis timeline	Pulse from the shell.
glacis backup --label pre-enum · glacis backups	Snapshot / list snapshots.
glacis restore <snapshot.json>	Restores as a NEW workspace — never overwrites.
glacis export --format md json txt html [-o file]	Export (html = styled report).

Exam-day checklist

First 15 minutes	Last 30 minutes
1. Launch glacis, pick theme (T, d). 2. Press E — Exam Mode badge (see below). 3. :ws init exam — clean workspace. 3. Import the scope/first scan (I). 4. :lhost auto · :lport 4444. 5. :m ejpt on the first target. 6. glacis backup --label start.	1. Station 0: any target without flags? NOW queue empty? 2. Walk every DEFERRED service once more. 3. Every question has a :q proof? Check count. 4. :export exam → answer from the dossier. 5. glacis export --format html -o exam.html. 6. glacis backup --label submitted.

AI-proctored exams (eJPT v2) — why this notebook is safe

Newer INE practicals are watched by an AI proctor looking for unauthorized help: AI assistants, chat, cloud services, anything leaving the exam environment. GLACIS is a personal notes file, and it stays provably on the right side of the line:

What the proctor sees	What GLACIS actually does
"Is an AI assisting you?"	No AI features exist. Playbooks and :ref are static text bundled with the app — a man page, not an advisor. Derived guidance ships off; leave it off.
"Is anything sent to the network?"	Zero network code paths. No telemetry, updates, sync, or sockets. Run glacis exam-check to audit the installed build yourself.
"Is it doing the exam for you?"	It cannot execute anything. It shows a command; you copy it and run it in your own terminal — same as your own cheat sheet.
"What is this window?"	Press E : the header shows a persistent EXAM MODE · OFFLINE NOTES badge — the window explains itself.

✓ Where your data lives

One SQLite file (default `~/ .local/share/glacis/notebook.db`, or `.glacis/` beside an initialised workspace) plus plain folders for scans and loot. Copy them to a USB stick and you have a full backup — or let **glacis backup** rotate snapshots for you. No network, ever: safe for exam NDAs by design. Always defer to your certifying body's current rules.